

Relative price dispersion and inflation regimes in South Africa

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Motivation

Motivation

Literature

Relative price dispersion through Lucas's aggregate supply curve

Empirical literaturer review

Data and methodology

Data

Our analysis uses monthly inflation data on 45 goods categories from January 2009 to November 2025 obtained from Statistics South Africa (Stats SA).¹ . **?@fig-headline_inflation** below shows the long term inflation rate in South Africa, and **?@fig-price_data** in the appendix shows the various price categories used in the analysis. **?@tbl-descriptive_stats** provides the descriptive statistics of the inflation rates for the various price categories used in the analysis.

¹The data is available at: <http://www.statssa.gov.za/> The data is seasonally adjusted.

Methodology

The relative price dispersion (RPD) is calculated in line with @lach1992 as the weighted standard deviation of individual price inflation rates from the aggregate inflation rate. The RPD is essentially the variability of relative prices of a given product. We first calculate the monthly inflation rate as follows:

$$\pi_{i,t} = \ln \left(\frac{P_{i,t}}{P_{i,t-1}} \right) * 100, \quad (1)$$

where $\pi_{i,t}$ is the inflation rate of good i at time t , $P_{i,t}$ is the price of good i at time t , and $P_{i,t-1}$ is the price of good i at time $t - 1$.

Methodology

Then, the RPD is calculated as:

$$RPD_t = \sqrt{\sum_{i=1}^n \omega_i (\pi_{i,t} - \pi_t)^2}, \quad (2)$$

where ω_j is the expenditure share of the j^{th} good, π_t is the headline or aggregate inflation rate at time t , and $\sum_{i=1}^n \omega_i = 1$. The expenditure weights (ω_i) used are derived from the Consumer Price Index (CPI) basket provided by Stats SA. Figure ?? shows the RPD calculated using the above formula. We use the 2025 CPI Stats SA weights. The weights are shown in **?@fig-weights** in the appendix.

Methodology

To examine the relationship between inflation and RPD, we estimate the following econometric model:

$$RPD_t = \beta_0 + \beta_1 |\pi_t| + \sum_{h=1}^3 \omega_h RPD_{t-h} + \mu_t, \quad (3)$$

where RPD_t is the relative price variance at time t , $|\pi_t|$ is the absolute value of aggregate inflation rate at time t , RPD_{t-h} are lagged values of RPD to account for its dynamic nature, and μ_t is the error term. The coefficient β_1 captures the effect of inflation on RPD. We include three lags of RPD based on Akaike Information Criterion (AIC) (and other tests) and to capture potential persistence in RPD. For robustness, we repeat this estimation using core inflation (excluding food and non-alcoholic beverages, and fuel and electricity) instead of headline inflation to account for the potential volatility in these components.

Methodology

The literature on RPD emphasises that the relationship between inflation and RPD maybe time-varying due to structural changes in the economy, monetary policy regimes, and other factors. To capture potential time variation in the relationship, we also estimate a rolling regression model with a 24-month and 60-month windows as follows:

$$RPD_t = \alpha_t + \beta_{1,t}|\pi_t| + \sum_{h=1}^3 \omega_{h,t}RPD_{t-h} + \mu_t, \quad (4)$$

where the coefficients α_t , $\beta_{1,t}$, and $\omega_{h,t}$ vary over time. This approach allows us to observe the stability of the coefficients over time, which is an indicator of non-linearity in the relationship between RPD and inflation.

Methodology

Next, we utilise a semi-parametric or partially linear approach to estimate the optimal level of inflation given the RPD. This approach allows us to model the relationship between inflation and RPD without imposing a strict functional form. Following the literature, we first specify a quadratic (U-shaped) relationship between inflation and RPD as follows:

$$RPD_t = \alpha + \pi_t + \beta_2 \pi_t^2 + \sum_{h=1}^3 \phi_h RPD_{t-h} + \epsilon_t, \quad (5)$$

where β_2 captures the curvature of the relationship between inflation and RPD. The optimal inflation rate that minimizes RPD can be derived by taking the first derivative of the equation with respect to π_t and setting it to zero, yielding:

$$\pi_t^* = -\frac{1}{2\beta_2}. \quad (6)$$

Methodology

To estimate this optimal inflation we implement a semi parametric approach as follows:

$$RPD_t = f(\pi_t) + \sum_{h=1}^3 \phi_h RPD_{t-h} + \epsilon_t, \quad (7)$$

where $f(\pi_t)$ is a non-linear smooth differentiable function of inflation and the RPD. Therefore, the optimal inflation rate can be obtained by finding the minimum of the estimated $f(\pi_t)$ function. $f(\pi_t)$ is estimated in two stages. First we estimate λ_h as follows:

$$RPD_t = \sum_{h=1}^3 \lambda_h \overline{RPD}_{t-h} + \eta_t, \quad (8)$$

where $\overline{RPD}_{t-h}$ are the residuals from a non-parametric regression of each lag of RPD_t on π_t . In the second stage, we estimate $f(\pi_t)$ non-parametrically as follows:

Methodology

To further disentangle the effects of RPD on inflation, we decompose headline inflation into its expected and unexpected components. As discussed in the literature section, for example, theory suggests that inflation uncertainty has a more pronounced effect on RPD due to information frictions faced by firms when adjusting prices [Lucas1973; Hercowitz1981; Cukierman1983].

We therefore estimate the following model:

$$RPD_t = \alpha + \beta_0 EIN_t + \beta_1 UIN_t + g(UIN_t) + \sum_{k=1}^3 \lambda_k RPD_{t-k} + \epsilon_t, \quad (10)$$

where EIN_t is the expected inflation rate, UIN_t is the unexpected inflation rate, and $g(UIN_t)$ is a non-linear function of unexpected inflation (UIN_t). We obtain the expected using **a simple moving average of headline inflation, and unexpected inflation as the standard deviation of inflation**. The estimation

Results

Results

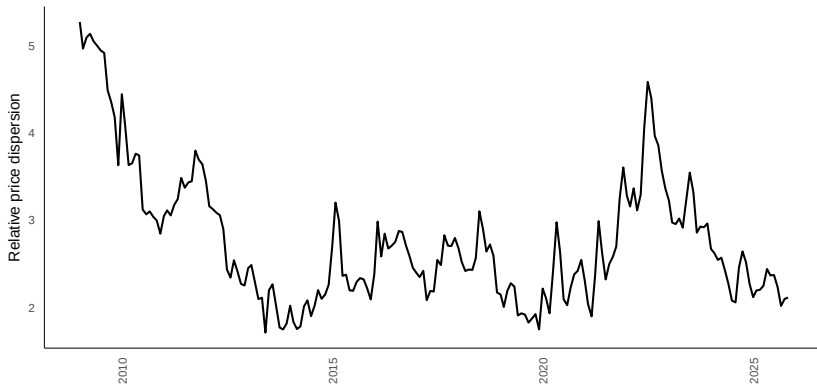


Figure 1: RPD over-time

Results

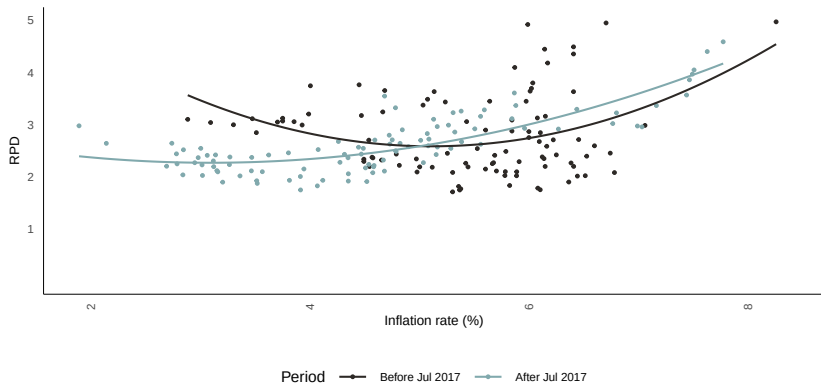


Figure 2: RPD vs headline Inflation

Results

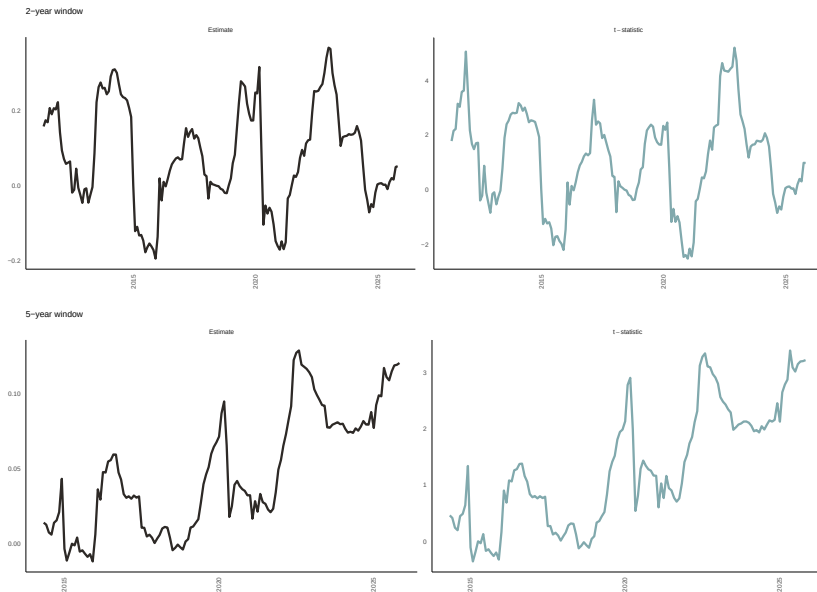


Figure 3: Rolling regressions. Note: t-statistics of greater (or less) than

Results

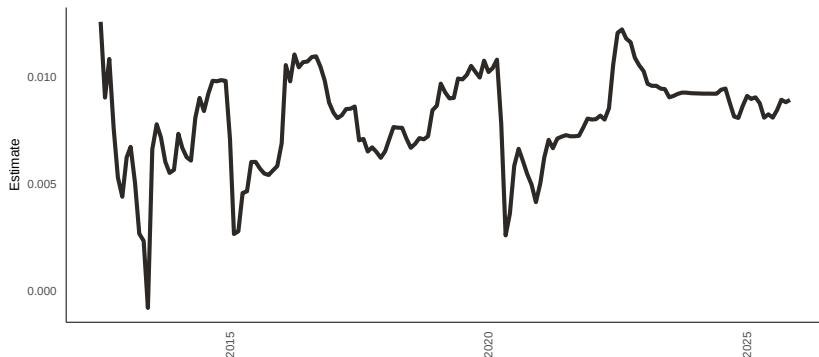


Figure 4: Recursive regressions. Note: Recursive regression estimates of the coefficient on absolute headline inflation with a fixed start date of 2012-07.

Results

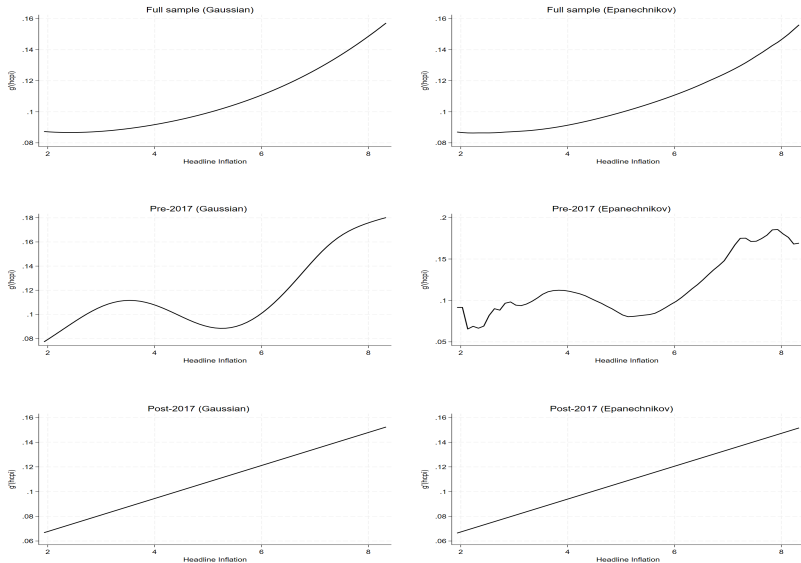


Figure 5: Semi-parametric regressions

Results

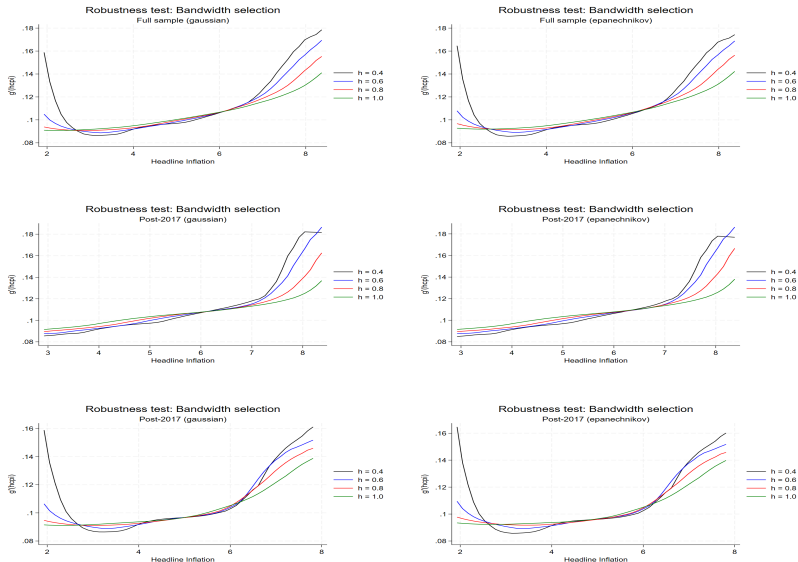


Figure 6: Semi-parametric regressions with alternative bandwidth

Results

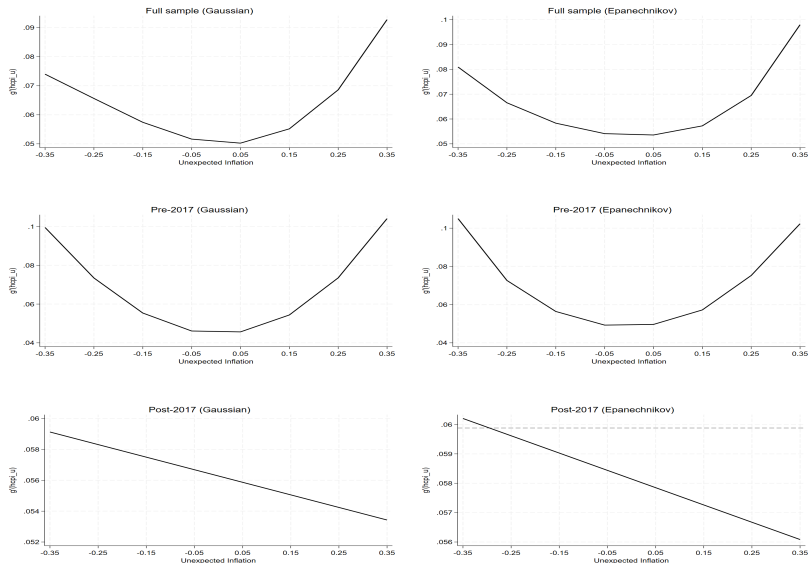


Figure 7: Semi-parametric regressions for unexpected inflation. Note:

Results

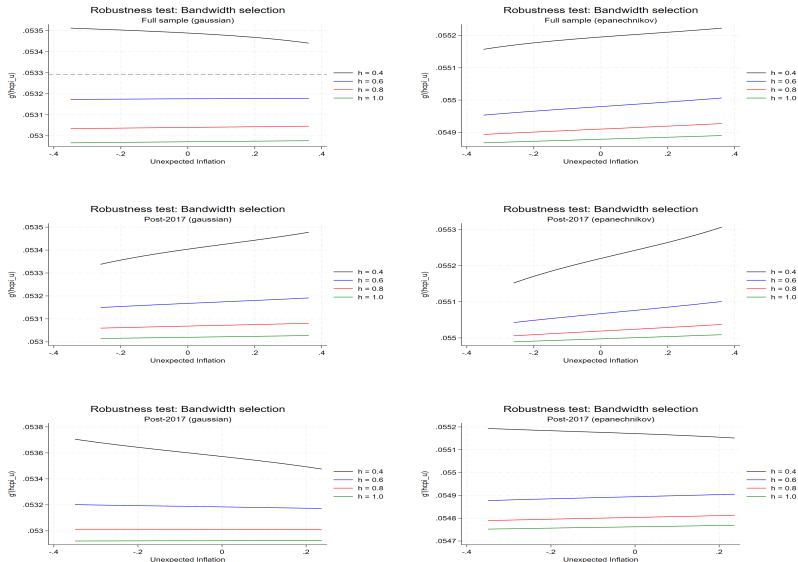


Figure 8: Semi-parametric regressions for unexpected inflation with

Results

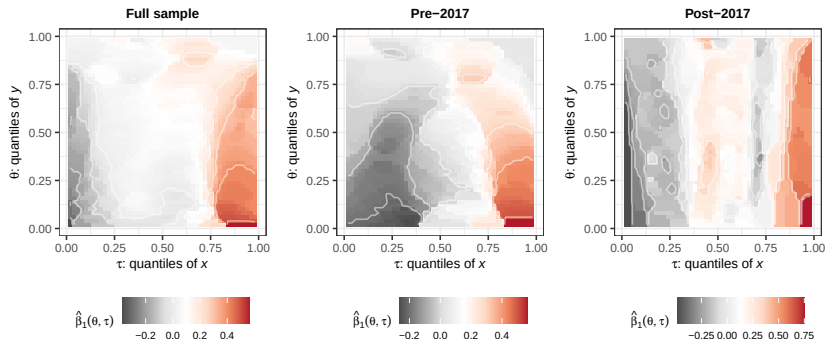


Figure 9: Quantile on quantile regressions for headline inflation. Note: In the quantile on quantile regression plots, the x-axis represents the quantiles of headline inflation, while the y-axis represents the quantiles of RPD. The color gradient indicates the strength and direction of the relationship between headline inflation and RPD across different quantiles. Warmer colors (e.g., red) indicate a stronger positive relationship, while cooler colors (e.g., blue) indicate a stronger negative relationship. The white areas represent quantiles where the relationship is not statistically significant.

Results

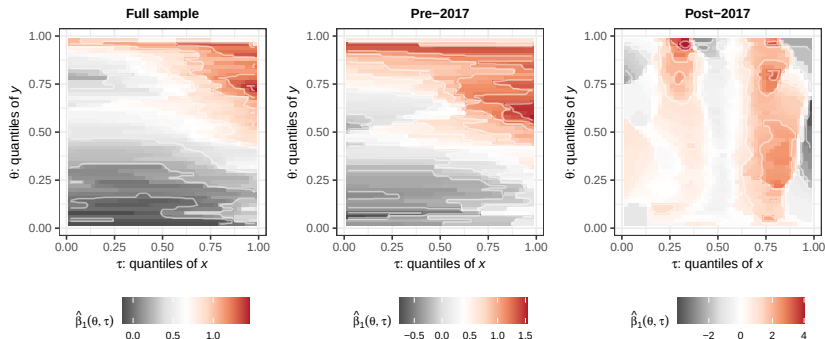


Figure 10: Quantile on quantile regressions for the unexpected component of inflation. Note: The x-axis represents the quantiles of the unexpected component of headline inflation, while the y-axis represents the quantiles of RPD.

Conclusions

Conclusion